

web technologies html css

this course will give you details of
web basics, html and css

client & server

1) **client** is a software which requests data.

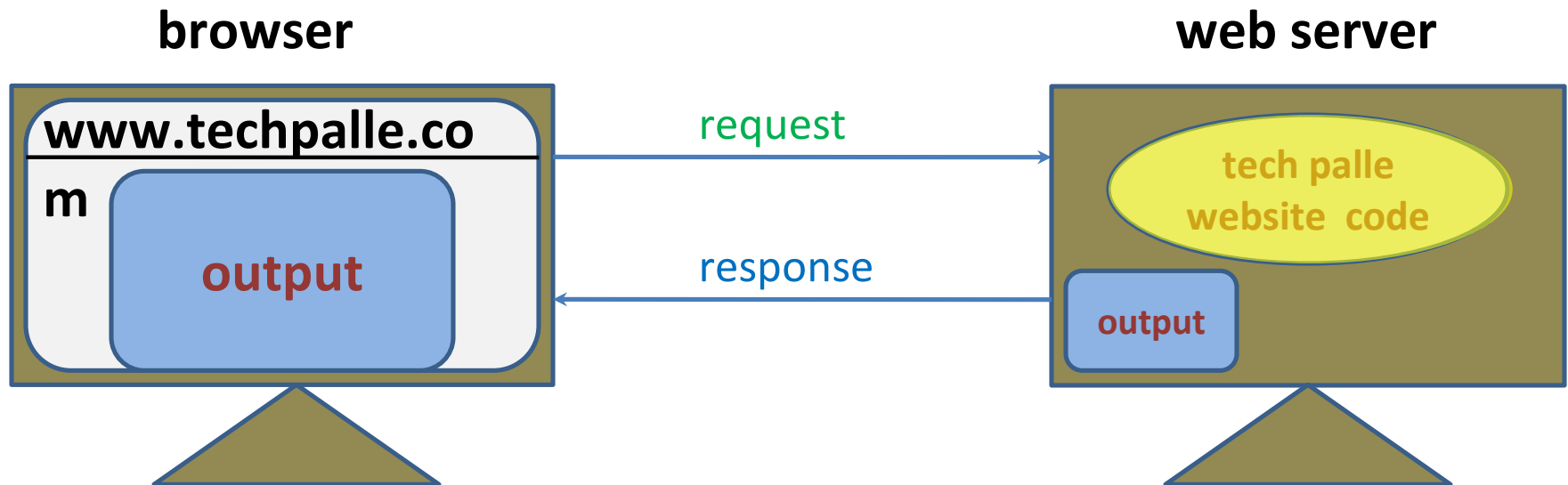
2) **server** is a software which gives data.

3) in web communication browser is **client**.

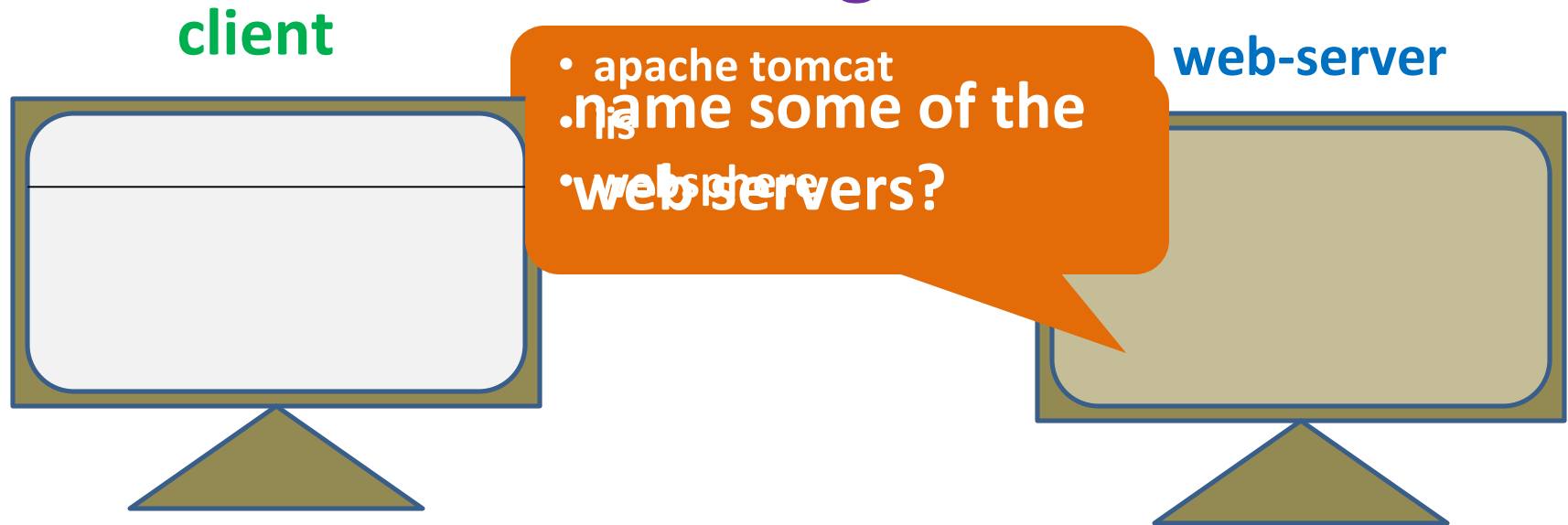
examples for client :google chrome ,internet explorer , etc

4) in web communication web server is **server**.

examples for server: iis , apache tomcat, etc



client and server side language technologies



what is client side language?

- languages which are executed by the **browser/client** are called as **client side language**
- html, css, java script etc.

what is server side language?

- languages executed by the **web-server** is called as **server side language**.
- c# , java, perl, python.

html

hyper text markup language

- it is not a **programming language** but it is a **presentation language**
- html is not **case sensitive**
- html file extension is **.html / .htm**
- html is **error free language**.

HTML file structure

```
<html>

  <head>
    <title>test page</title>

  </head>
  <body>

  </body>
</html>
```

js code. styles code.

html code + js code + styles code + plain text

Below are not supported in
HTML

Datatypes , variables

Loops, conditional statements

creating an html file

- different tools for creating html files
 - eclipse
 - notepad
 - notepad++
 - dream viewer
 - visual studio
 - visual studio (web developer)
 - pycharm

**trainer: show how to open
html page in visual studio by
using web application**

 and

-
 -> is used to insert a line break

eg: pale
technologies

Output is:



Pale
technologies

 [?] non breaking space used to give specific space between the word

eg: pale technologies

Output is:



Pale technologies

frequently used tags

headings tag: <h1></h1>(font size is more),<h2>,<h3>,<h4>,<h5>,<h6>(font size is least)

we can control font size using styles

graphical user interface controls:

text box:

some data

```
<input type="text" id="tbeid"/>
```

tag / element name

property / a property / attribute

key

val

key

value

radio button:

Gender: ☐ MALE ☐ FEMALE ☐ OTHERS

```
Gender: <input type="radio" id="rdmale" name="gender"/> MALE  
<input type="radio" id="rdfemale" name="gender"/> FEMALE  
<input type="radio" id="rdothers" name="gender"/> OTHERS
```

check box:

Courses: ☒ PYTHON ☐ JAVA ☒ WEB TECHNOLOGIES

```
courses: <input type="checkbox" id="ckpython"/> PYTHON  
<input type="checkbox" id="ckjava"/> JAVA  
<input type="checkbox" id="ckhtml"/> WEB TECHNOLOGIES
```

drop down list:

STATES:



STATES : `<select id="ddlstates">`

`<option>--select--</option>`

`<option>Karnataka</option>`

`<option>Kerala</option>`

`<option>Andhra pradesh</option>`

`</select>`



	
--select--	
Karnataka	
Kerala	
Andhra pradesh	

password
:

```
<input type="password" id="tbpwd"/>
```

email:

abc@gmail.com

```
<input type="email" id="tbemail"/>
```

button:

submit button will submit entire form data to server

SUBMIT

```
<input type="submit" id="btnsubmit" value="SUBMIT"/>
```

hello

```
<input type="button" id="btnhello" value="hello"/>
```

button will not send the entire page data to server

anchor tag: signin

hyperlink

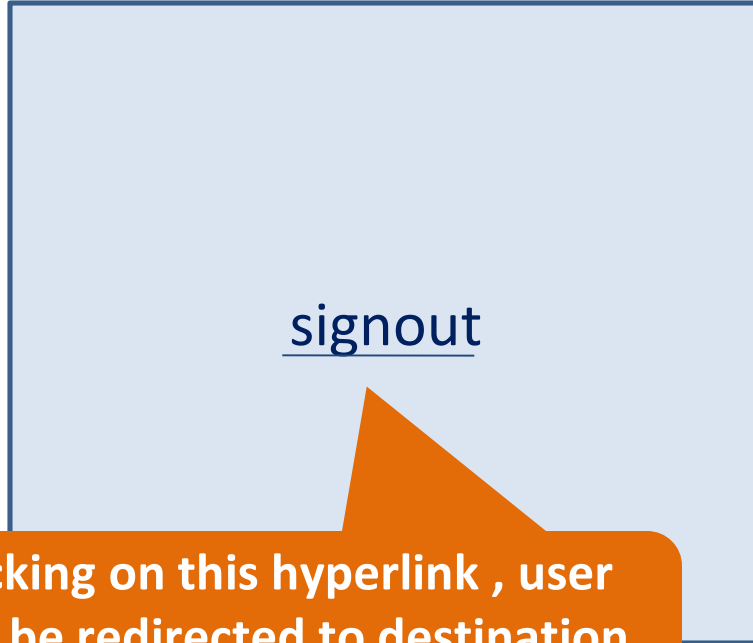
```
<a href="signin.html">signin</a>
```

user will be redirected to

browser will create hyperlink with signin

anchor tag: `<a href="signinout.html">signinout</a>`

home.html



clicking on this hyperlink , user
will be redirected to destination
page

signinout.html



destination page

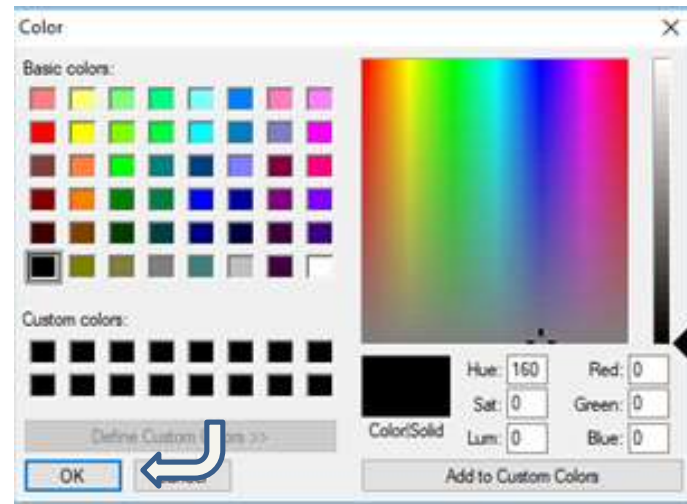
create an image: ``

input type="color"

This will allow you to select a color from color picker

select your favourite color:<input type="color" />

Select your favorite color:



input type="number"

we can limit the range of permissible input values

```
Number<input type="number" min="4" max="10"/>
```

user can give numbers from 4 to 10

Number

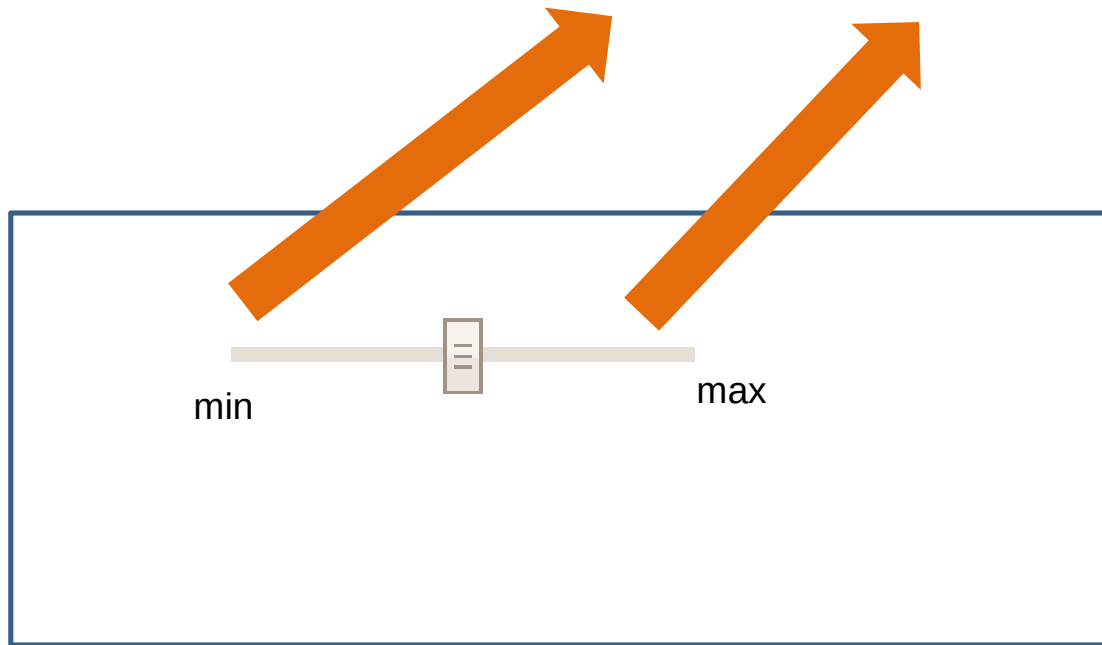
Value must be lesser than or equal to 10

note: when you give **invalid** number you will get a error message

input type="range"

`type="Range"` will allow us to create a slider in the webpage

```
<input type="range" min="5" max="11" step="2" />
```



input type="date" input

Monthname

Year

```
<input type="Date"/>
```

```
<input type="month" />
```

dd-----yyyy

February, 2018

Sun	Mon	Tue	Wed	Thu	Fri	Sat
28	29	30	31	1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	1	2	3

-----, ----

February, 2018

Sun	Mon	Tue	Wed	Thu	Fri	Sat
28	29	30	31	1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	1	2	3

input type="email"

this will allow only the email addresses

```
<input type="Email"/>
```

when the email is
typed in proper
format we wont get
error

palle

Please insert an '@', palle is missing '@'

Note:

if the email format is not correct you will get this error

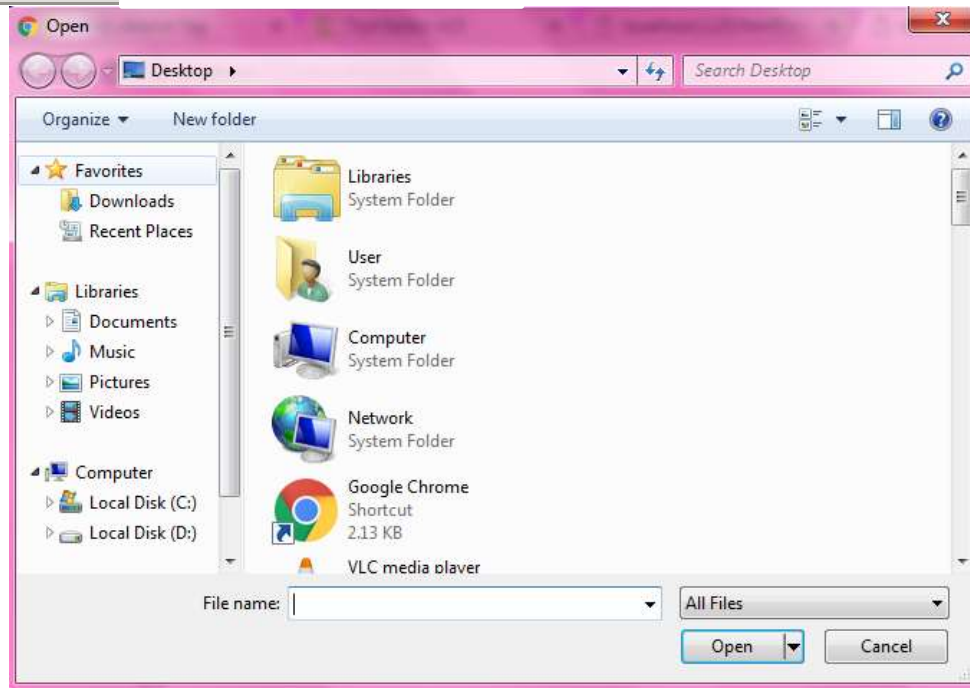
input type="file"

File type will allow the user to choose one or more files from their device storage.

```
<input type="file" multiple/>
```

Choose File

3 files



<details> element

The **<details> tag** specifies additional **details** that the user can view or hide on demand

```
<details>  
  <summary>PALLE</summary>  
  Palle started in 2010,providing job guaranteed IT training to individuals  
</details>
```

▼ PALLE

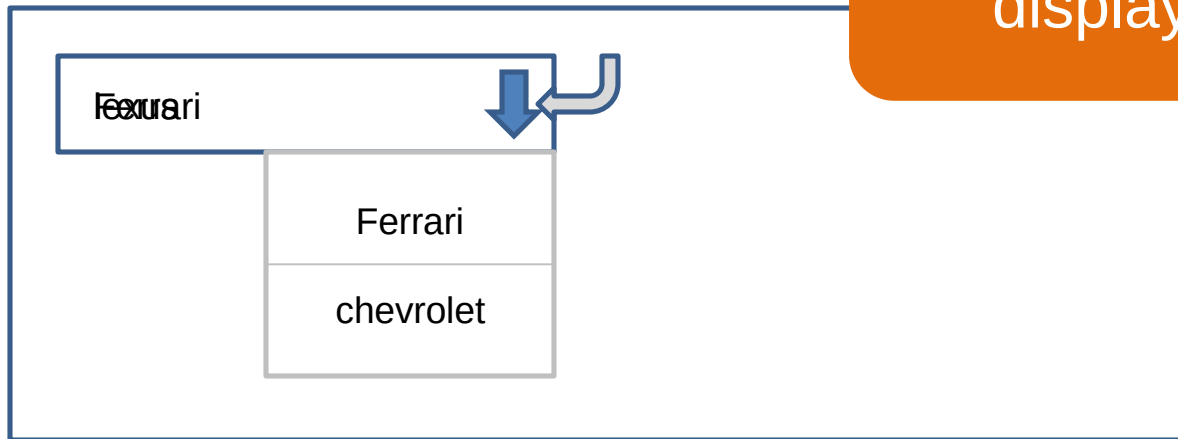
Palle started in 2010 ,providing Job Guaranteed IT Training to Individuals

<datalist>

when **datalist** **id** attached to a **textbox** the text box will work like a dropdown list .

```
<datalist id="car">
<option value="ferrari"></option>
<option value="bentley"></option>
</datalist>
<input list="car" />
```

datalist alone will not display any UI



validations

using validations we can
check the users data format
correctness

required minlength
min max maxlength
pattern

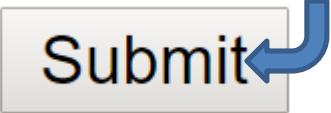
required attribute

required attribute will not have value. it is used for checking **emptiness** of a textbox before submitting the form.

example :-

```
Name <input type="Text" required>
```

Name

 Submit

 Please fill out this field.

placeholder

1. The **placeholder** attribute is **used** in **input tags**.
2. It will **show** a **short hint** that **describes** the **expected value**

example :-

```
Name:<input type="Text" placeholder="Enter your Name" />
```



Name: Enter your Name

minlength and maxlength attribute

It will specify the minimum and maximum length of the text data (strings)

Example:

Enter the name: `<input type="text" id="tb1" minlength="3" maxlength="10" />`

Enter the name:

You cannot type more than 10 characters (maxlength) in this textbox

min and max attribute

It specifies the minimum and maximum values of numerical input types.

Example:

Enter any number: `<input type="number" id="tb1" min="2" max="20"/>`

Enter any number:

Value must be less than or equal to 20.

This textbox will allow to enter the numbers between the min and max values

creating html gui sample page

Eid:

Name:

Gender: ☐ Male ☐ Female ☐ Others

BG: ☐ O+ve ☐ O-ve ☐ b+ve

States:

----select----

KA

AP

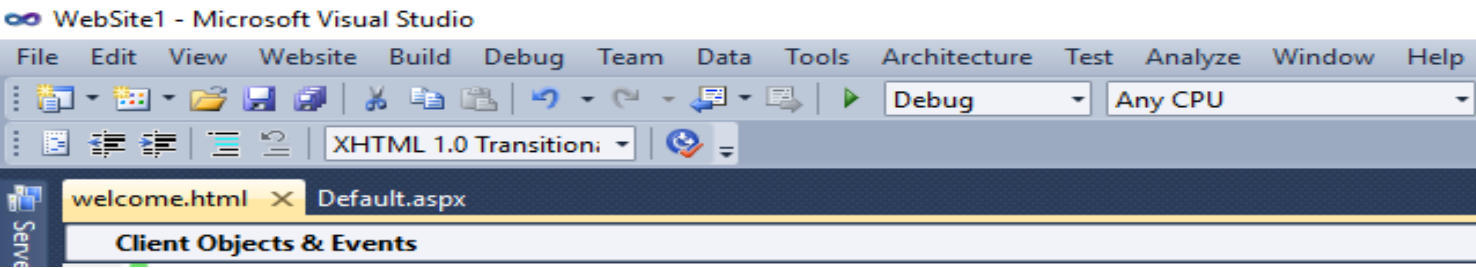
When to use TextBox
control?

When to use Radio ?

When to use CheckBox?

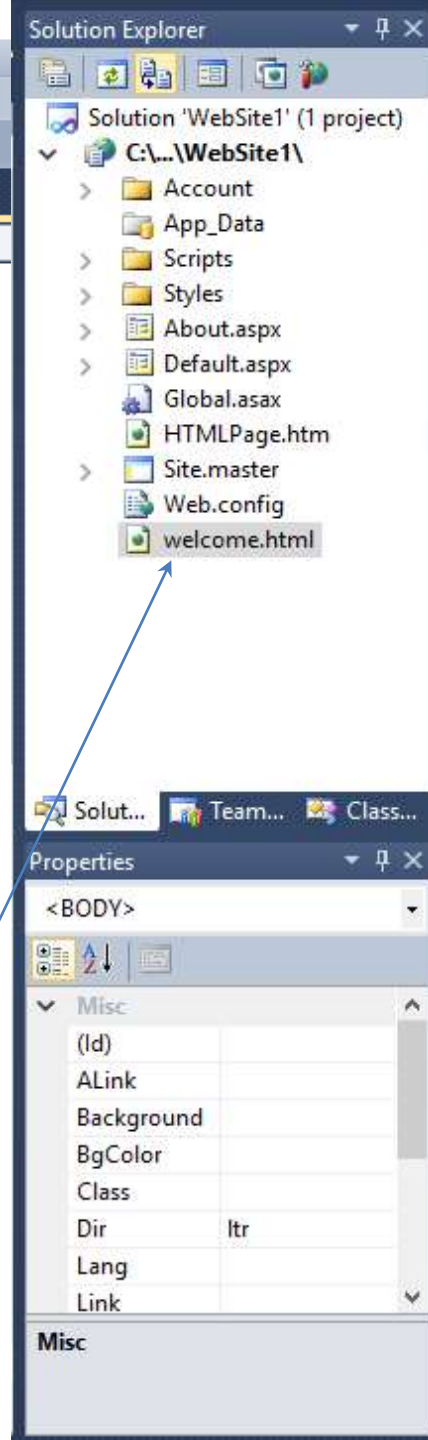
When to use DropDownList?

When to use Submit Control?



```
<html>
<head>
<title> EMPLOYEE REGISTRATION</title>
</head>
<body>
EID:<input type="text" id="tbeid"/><br />
NAME:<input type="text" id="tbname"/><br/>
GENDER:<input type="radio" id="rdmale"/>MALE
      <input type="radio" id="rdfemale"/>FEMALE
      <input type="radio" id="rdothers"/>others<br />
BG:<input type="checkbox" id="chko+ve"/>O+VE
   <input type="checkbox" id="chko_ve"/>o-ve
STATES:<select id="ddlstates">
      <option>--select--</option>
      <option>-karnataka</option>
      <option>-kerala</option>
      <option>andrapradesh</option>
    </select>
<input type="submit" id="btnsubmit" value="submit"/>
<input type="button" name="name" value="button" />
</body>
</html>
```

Right click and select view in browser



View in browser output



EID:
NAME:
GENDER: ☐ MALE ☐ FEMALE ☐ others
BG: ☐ O+VE ☐ O+VE
STATES:

html tables

tags used to create tables in html

<table> → Table tag used to create tables in html

<tr> → tr tag used to create the rows in html tables

<th> → th tag used to give column names of our html tables

<td> → td tag used to insert table data in html

create the following table in html??

```
<table border="1" cellspacing="0">
<tr>
  <th> Pid </th>
  <th> Name </th>
  <th> Age </th>
</tr>
<tr>
  <td> 101 </td>
  <td> Ravi </td>
  <td> 72 </td>
</tr>
<tr>
  <td> 102 </td>
  <td> Paul </td>
  <td> 70 </td>
</tr>
</table>
```

Pid	Name	Age
101	Ravi	72
102	Paul	70

pid	Name	Age
101	Ravi	72
102	Paul	70

req: write html code for creating the following table

Employee.html

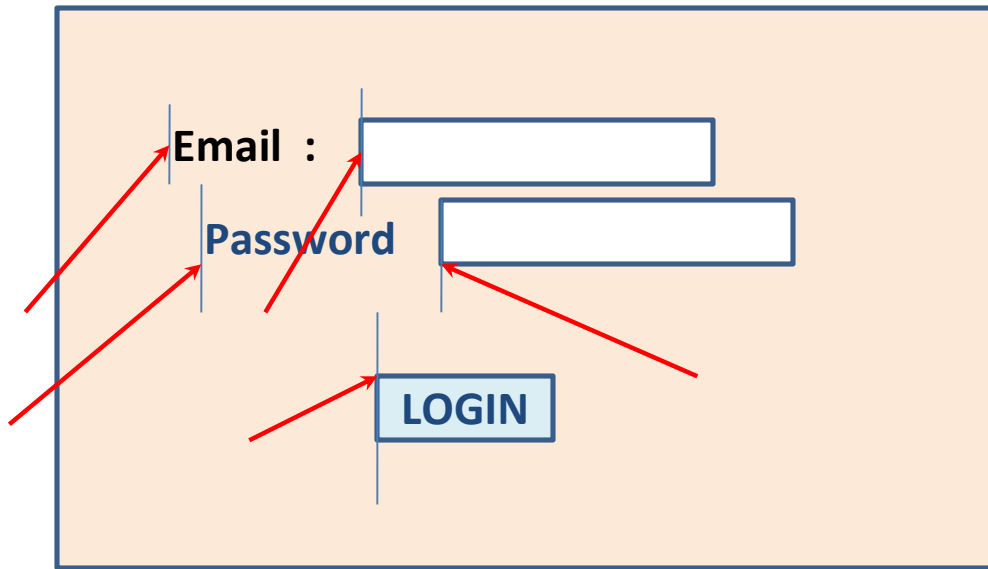
Eid	Name	Salary
101	Ravi	82000
102	Paul	90000
103	Bill	98000

Eid	Name	Salary
101	Ravi	82000
102	Paul	90000
103	Bill	98000

```
<table border="1" cellspacing="0">
  <tr>
    <th>Eid</th>
    <th>Name</th>
    <th>Salary</th>
  </tr>
  <tr>
    <td>101</td>
    <td>Ravi</td>
    <td>82000</td>
  </tr>
```

```
<tr>
  <td>102</td>
  <td>Paul</td>
  <td>90000</td>
</tr>
<tr>
  <td>103</td>
  <td>Bill</td>
  <td>98000</td>
</tr>
</table>
```

userinterface alignment



Incorrect Alignment

When the user observe this User Interface they will not get interest in using website

➡ Now will understand how to give proper alignment to User Interface

Html code for this user interface



```
<html>
<head></head>
<body>
  <table>
    <tr>
      <td>Email : </td>
      <td><input type="text" id="tbemail" /></td>
    </tr>
    <tr>
      <td>Password : </td>
      <td><input type="password" id="tbpassword" /></td>
    </tr>
    <tr>
      <td> </td>
      <td><input type="button" id="btnLogin" value="LOGIN" /></td>
    </tr>
  </table>
</body>
</html>
```

UI with Proper Alignment



Email :	
Password :	
	LOGIN

Email :	
Password :	
	LOGIN

USER INTERFACE ALIGNMENT LAB

Eid:

Name:

Gender: ☐ Male ☐ Female ☐ Others

BG: ☐ O+ve ☐ O-ve ☐ b+ve

states:

Eid:

Name:

Gender: ☐ Male ☐ Female ☐ Others

BG: ☐ O+ve ☐ O-ve ☐ b+ve

States:

KA

AP



Execute all code and do
all Exercises at Home

End of
DAY 1

CSS

- CSS is a **styling language** (Its not a programming language or not a markup language)
- CSS stands for **cascading Style sheet**
- we can apply styles to webpages or to html elements
- CSS selectors are **case sensitive**
- File extension is **.css**

styles without css

req: make the background color of html page 'yellow'

we will write

`<body style="background-color:Yellow">`

Eid:

Name:

Gender: ☐ Male ☐ Female ☐ Others

BG: ☐ O+ve ☐ O-ve ☐ b+ve

States: 

background color
for html page
changed to yellow

most of the web designers will be specifying color in **hexadecimal** format or hexadecimal code

applying styles using hexadecimal

Named colors are **limited** (ex.. Like white, blue, black etcc...)

Using hexadecimal representation we can produce **unlimited** colors



Hexadecimal code Represented by **#XXXXXX**

- **#** followed by **6** character
- Here X can be any character between a-f or 0-9

Applying yellow color using **Hexadecimal**

```
<body style="background-color:#ffff00">
```

Hexadecimal value for yellow color

Eid:

Name:

Gender: ☐ Male ☐ Female ☐ Others

BG: ☐ O+ve ☐ O-ve ☐ b+ve

States: ▼

Background color for html page changed to yellow color

Hexadecimal value for white → **#ffffff**

Hexadecimal value for black → **#000000**

applying background color and font color

Req :

Eid:

Name:

Background color **red** and
font color of textboxes
white.

```
<tr>
  <td>Eid:</td>
  <td><input type="text" id="tbEid" style="background-color:red;color:White"> </td>
</tr>
<tr>
  <td>Name:</td>
  <td><input type="text" id="tbName" style="background-color:red;color:White"></td>
</tr>
```

background color & font color

Req :

States:



Change Background color and text color of dropdownlist.

```
<select id="ddlStates" style="background-color:Orange;color:White">
```

Key

Value

note

you are allowed to apply any number of styles, and if you are using multiple styles you have to use semicolon(;) to separate each styles

problem with style property

Eid: When same style applied to multiple controls
Name: using style property

```
<td><input type="text" id="tbEid" style="background-color:red"/> </td>  
<td> <input type="text" id="tbName" style="background-color:red"></td>
```

This will leads to code duplication, this approach is called **Inline approach**

➡ we can eliminate this problem by using CSS

css selectors

Types of selectors

- **Class** selector
- **Id** selector
- **Element** selector
- **Elements with attribute** selectors
- ~ selector
- + selector

Note: All CSS selectors code should be written in **.CSS** file.

class selector

Syntax:

```
.classname  
{  
-----  
Styles code  
-----  
}
```

Name: `<input type="text" id="tbname" value="Palle">`

Name:

Apply these styles using **Class** Selector:

Name: `<input type="text" id="tbname" value="Palle" class="Ab" >`

Name:

xyz.css

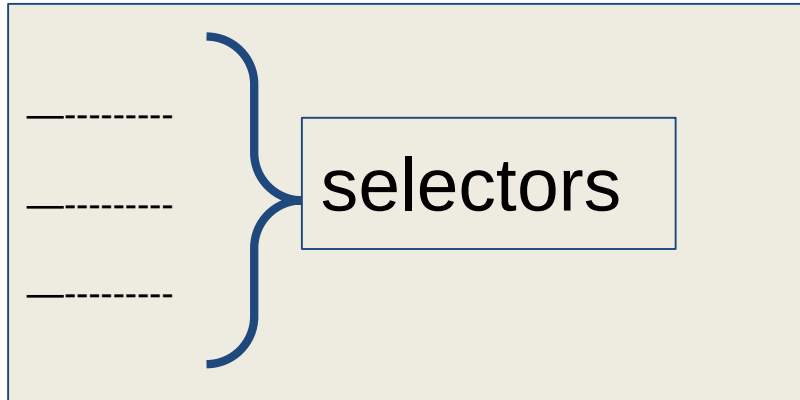
```
.Ab {  
    background-color: Orange;  
    color: Blue  
}
```

how to apply this style to control ?

link .css file with html

linking css file with html page

xyz.css



Req: I would like to use xyz.css styles in A.html

how to use?

link css file with html in head section

how to link?

A.html

```
<html>
<head>
  <link rel="stylesheet" href="xyz.css">
</head>
<body>

</body>
</html>
```

id selector

Syntax:

```
#html_element_id  
{  
-----  
Styles code  
-----  
}
```

Sample:

```
#tbage {  
    background-color: Red;  
    color: Aqua  
}
```

Age:

Age:

Requirement:-

Age:

applying same styles on multiple control using id selector

Cell: `<input type="text" id="tbcell" value="9875412496">`

States: `<select id="ddlstates">`

`<option> TN </option>`

`<option> KA </option>`

`<option> AP </option>`

`</select>`

Cell:

9875412496

States:

TN



`#tbcell`

`{`

`background-color: Green;`

`color: White`

`}`

`#ddlstates`

`{`

`background-color: Green;`

`color: White`

`}`



Apply these styles:

Cell:

9875412496

States:

TN



`#tbcell, #ddlstates`

`{`

`background-color: Green;`

`color: White`

`}`

element selector

Syntax:

```
Element_name  
{  
-----  
Styles code  
-----  
}
```

```
input  
{  
    background-color: Brown;  
    color: White  
}
```

What is the
Element name?

Name: `<input type="text" id="tbname" value="Palle">`

Name:

Apply these styles using Element Selector:

Name:

applying same styles on multiple control using element selector

```
Cell: <input type="text" id="tbcell" value="9875412496">
States: <select id="ddlstates">
  <option> TN </option>
  <option> KA </option>
  <option> AP </option>
</select>
```

Cell:

States:



```
input, select
{
  background-color: Maroon;
  color: Aqua
}
```

Apply these styles using Element Selector:

Cell:

States:



elements with attribute selectors

Syntax:

```
element name[attributename=value]
{
  -----stylecode-----
}
```

```
input[type = text]
{
  background-color: Blue;
  color: Red
}
```

Example:

```
Name: <input type="text" id="tbname" value="Lakshith">
Age: <input type="text" id="tbage" value="11">
DOB: <input type="text" id="tbdob" value="11-02-2009">
```

Apply these styles using element with attribute selectors:

Name:

Age:

DOB:

Name:

Age:

DOB:

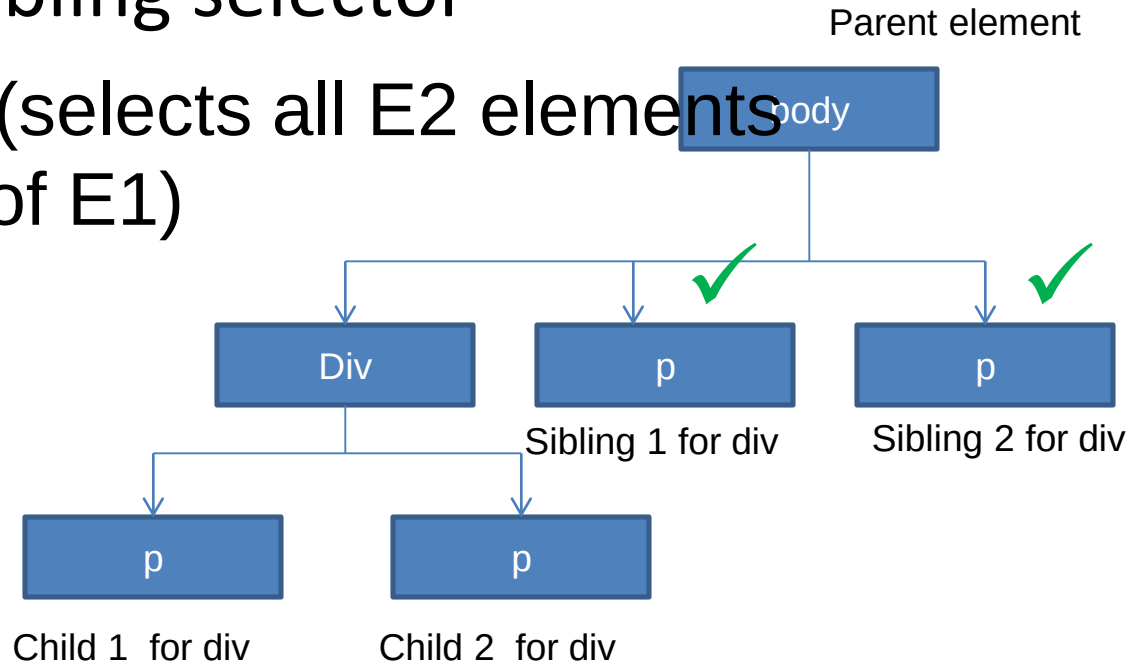
~ selector

~ is a general sibling selector

Syntax: E1 ~ E2 (selects all E2 elements which are siblings of E1)

Example:

```
div ~ p
{
  style;
}
```



It will select all the <p> tags which are present after the <div> tag.

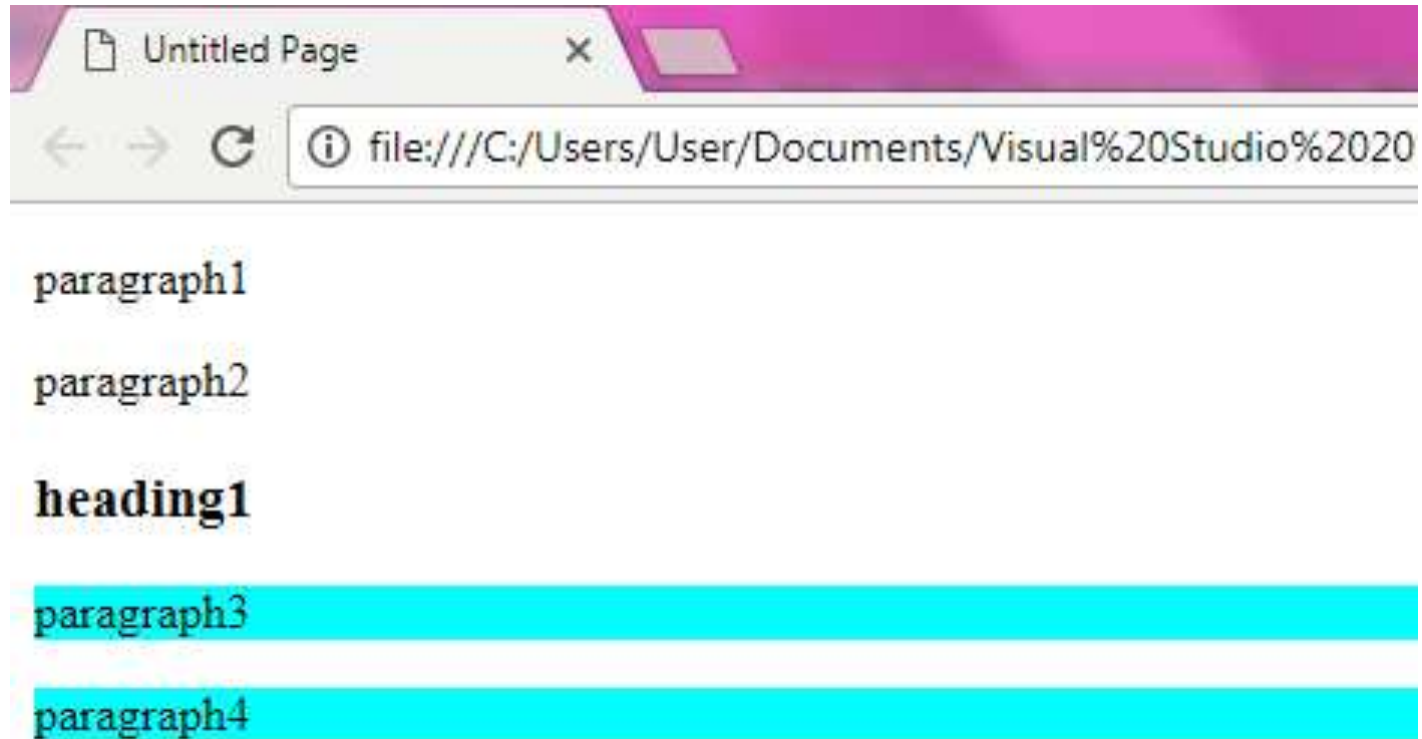
~ selector example

```
<body>  
<div>  
<p>paragraph1</p>  
<p>paragraph2</p>  
</div>  
  
<h3>heading1</h3>  
<p>paragraph3 </p>  
<p>paragraph4</p>  
  
</body>
```



```
div ~ p  
{  
  background-color:Aqua;  
  color:Black;  
}
```

~ output

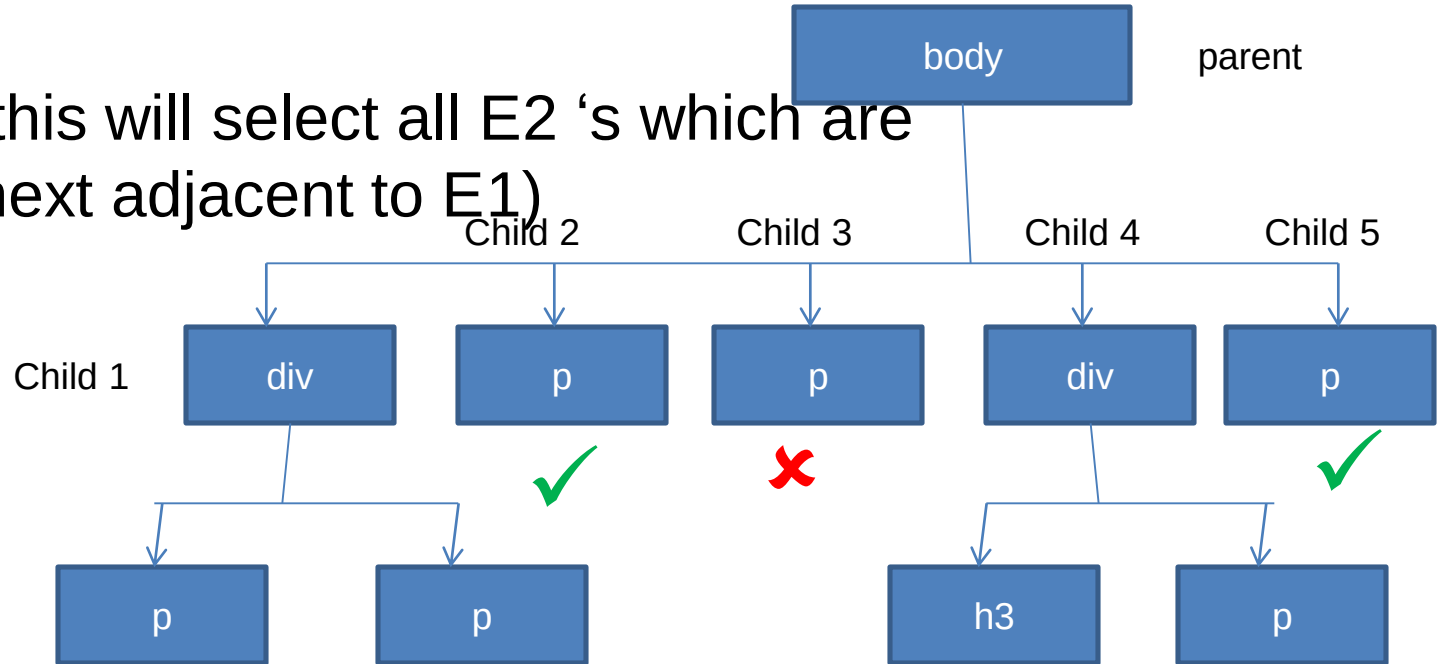


+ selector

+ is an immediate adjacent sibling selector

Syntax:

E1 + E2 (this will select all E2 's which are immediately next adjacent to E1)



Example:

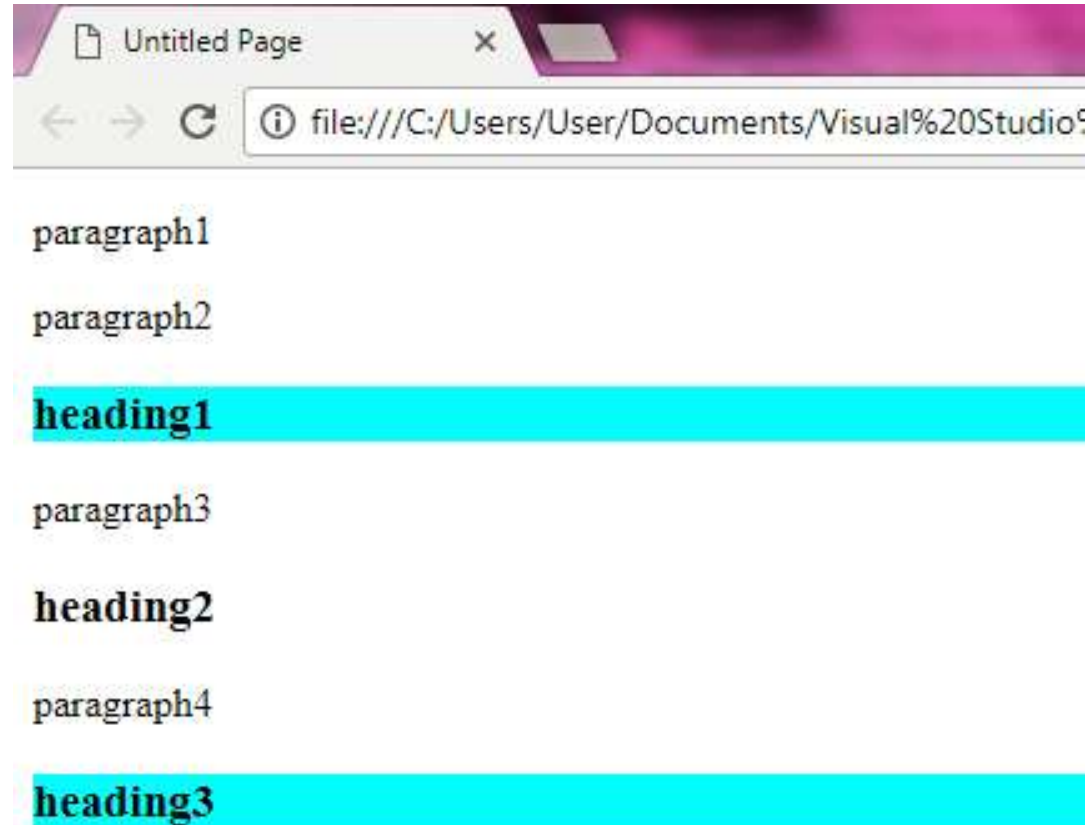
```
div + p
{
  Style code;
}
```

Will select the only <p> tags which are immediately present after a <div> tag

+ selector example

```
<body>
<div>
<p>paragraph1</p>
<p>paragraph2</p>
</div>
<h3>heading1</h3>
<p>paragraph3</p>
<h3>heading2</h3>
<p>paragraph4</p>
<div></div>
<h3>heading3</h3>
</body>
```

```
div + h3
{
    background-
color:Aqua;
    color:Black;
}
```



> Selector

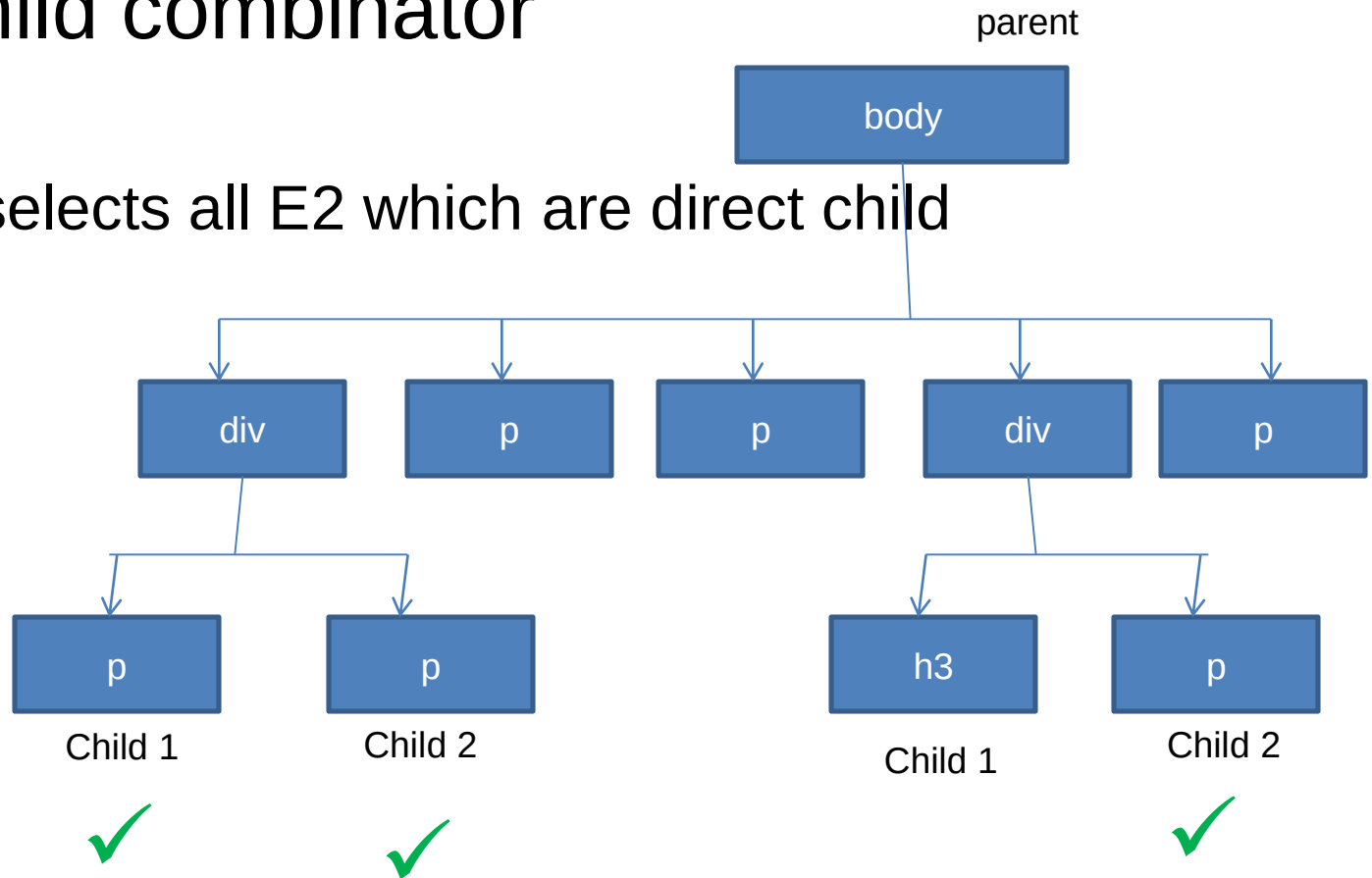
> Is a child combinator

Syntax:

E1 > E2 (selects all E2 which are direct child
For E1)

pattern

```
div > p  
{  
  Style code;  
}
```



Will select all the direct <p> child of <div> tags

> Selector example

```
<body>
```

```
<div>
```

```
<p>paragraph1</p>
```

```
<p>paragraph2</p>
```

```
</div>
```

```
<h3>heading1</h3>
```

```
<p>paragraph3</p><h3>hea
```

```
</body>
```

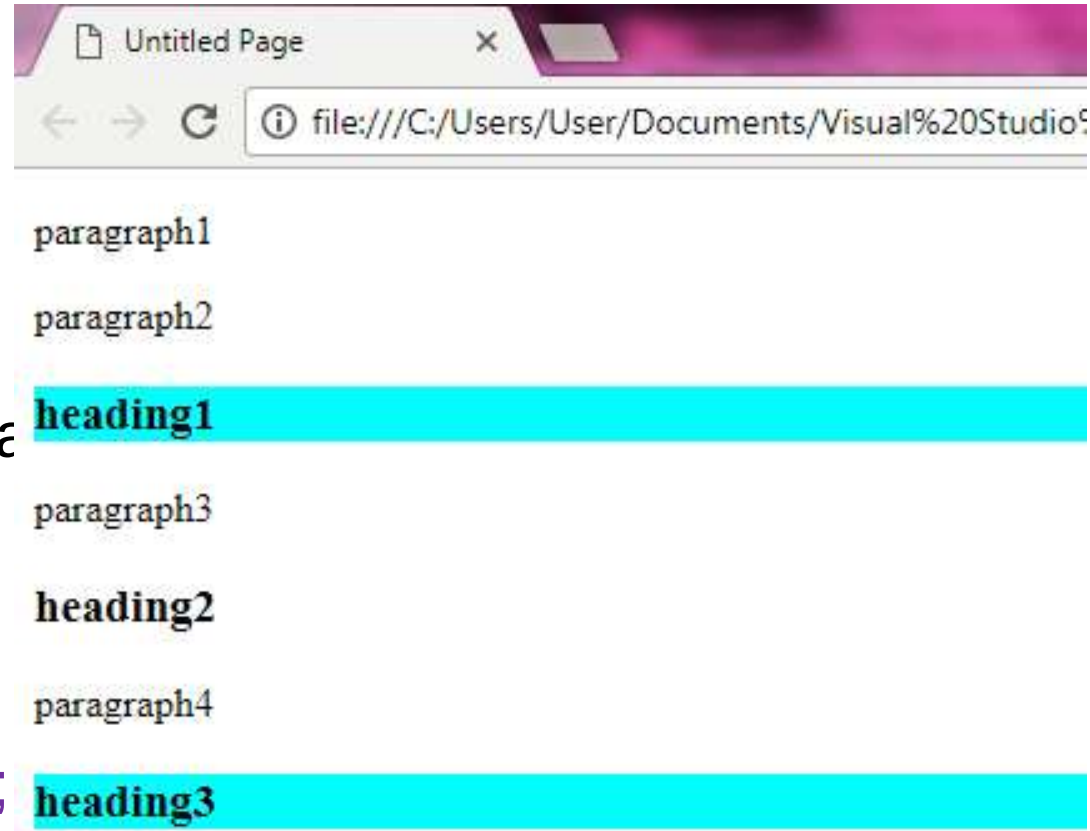
```
div>p
```

```
{
```

```
background-color:Aqua;
```

```
color:Black;
```

```
}
```



Single space selector

It is a descendant combinator

Syntax:

E1 E2 (selects all the direct and indirect E2 type elements which are children for E1)

pattern

div li

{

Style code;

}

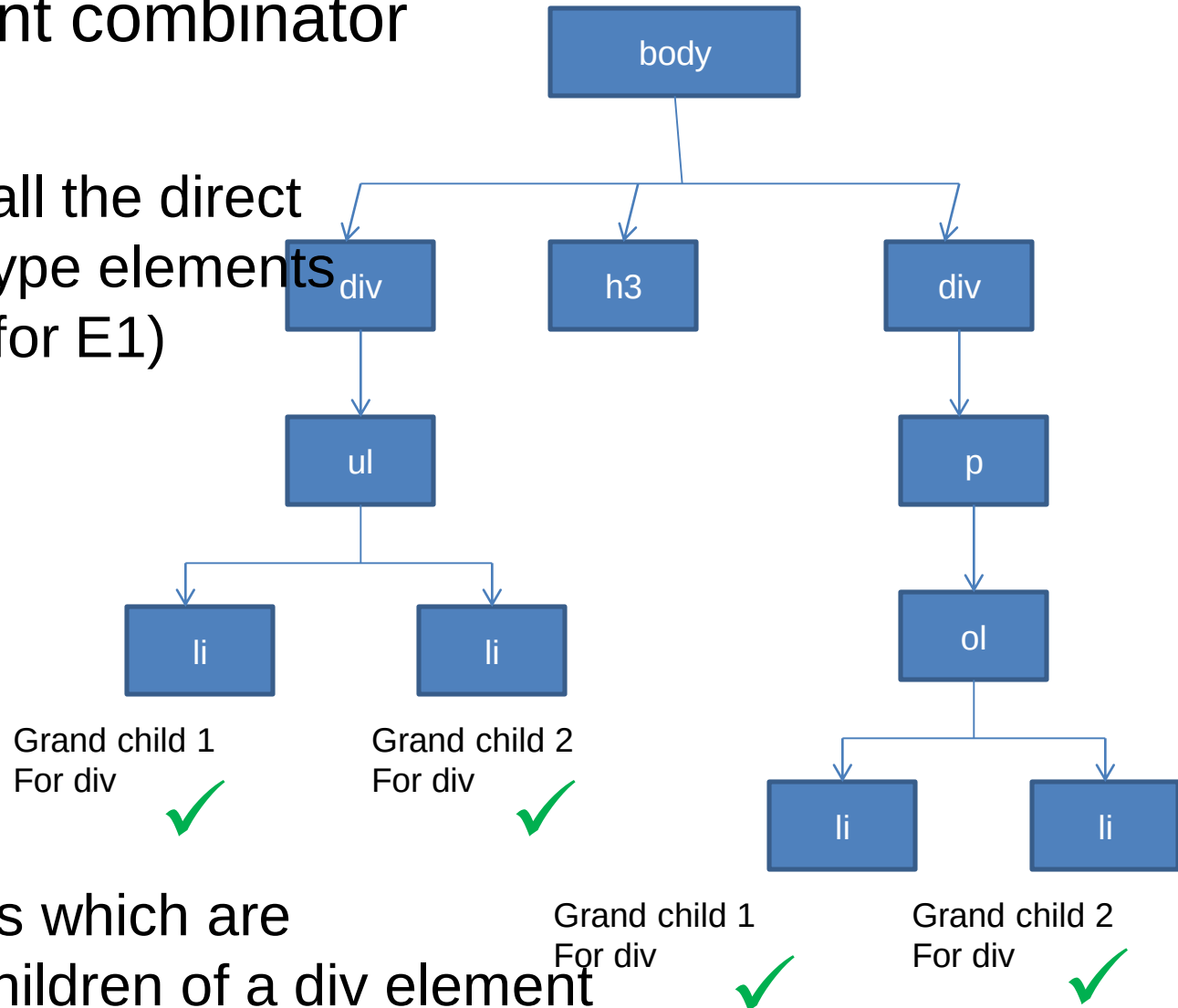
Grand child 1
For div ✓

Grand child 2
For div ✓

Grand child 1
For div ✓

Grand child 2
For div ✓

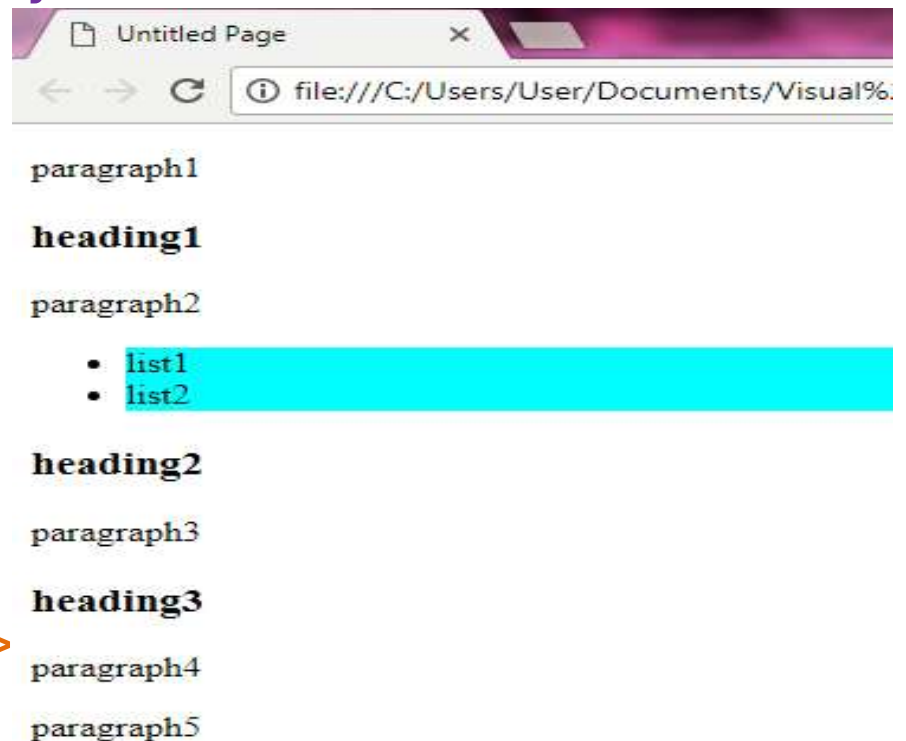
Will select the tags which are descendant / grand children of a div element



Single space with example

```
<body>
<div>
<p>paragraph1</p>
<h3>heading1</h3>
<p>paragraph2</p>
<ul>
  <li>list1</li>
  <li>list2</li>
</ul>
</div>
<h3>heading2</h3>
<p>paragraph3</p>
<h3>heading3</h3>
<p>paragraph4</p>
<div><p>paragraph5</p></div>
</body>
```

```
div li
{
  background-color:Aqua;
  color:Black;
}
```



* Selector

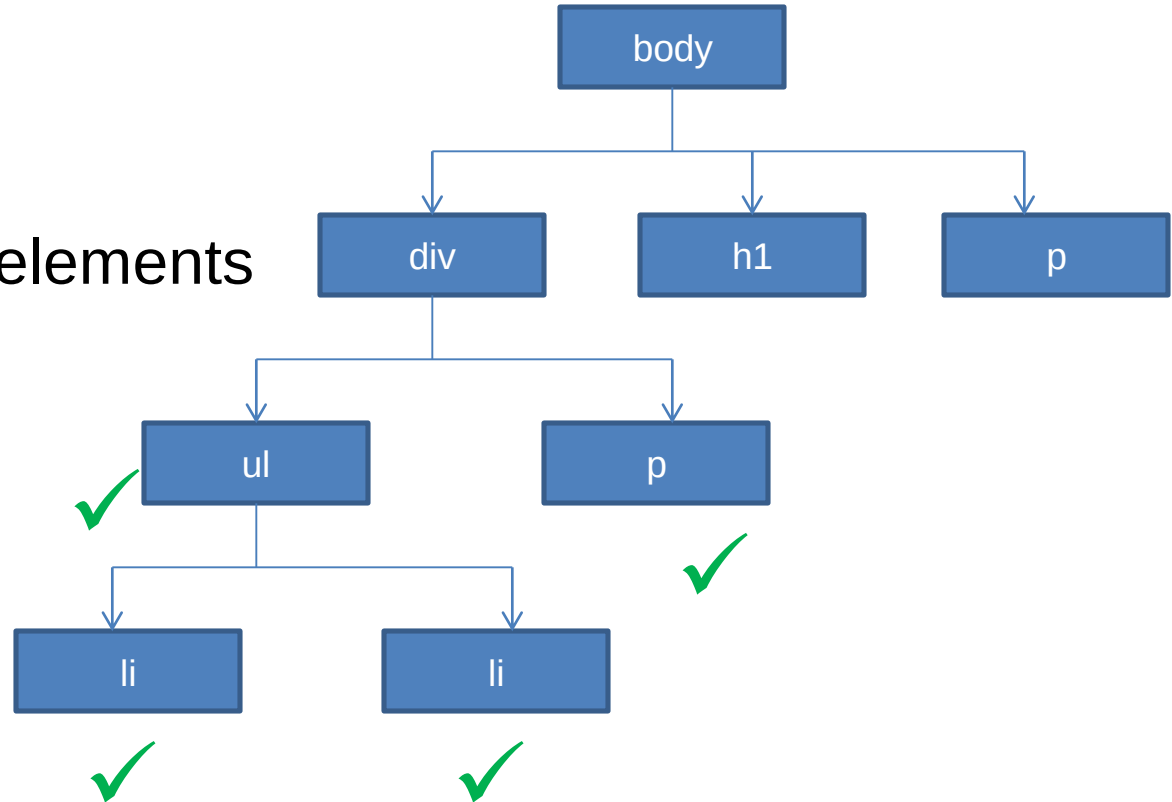
* Is a universal selector

Syntax:

E1 * (selects all elements
Present inside E1)

pattern

```
div *  
{  
  Style code;  
}
```

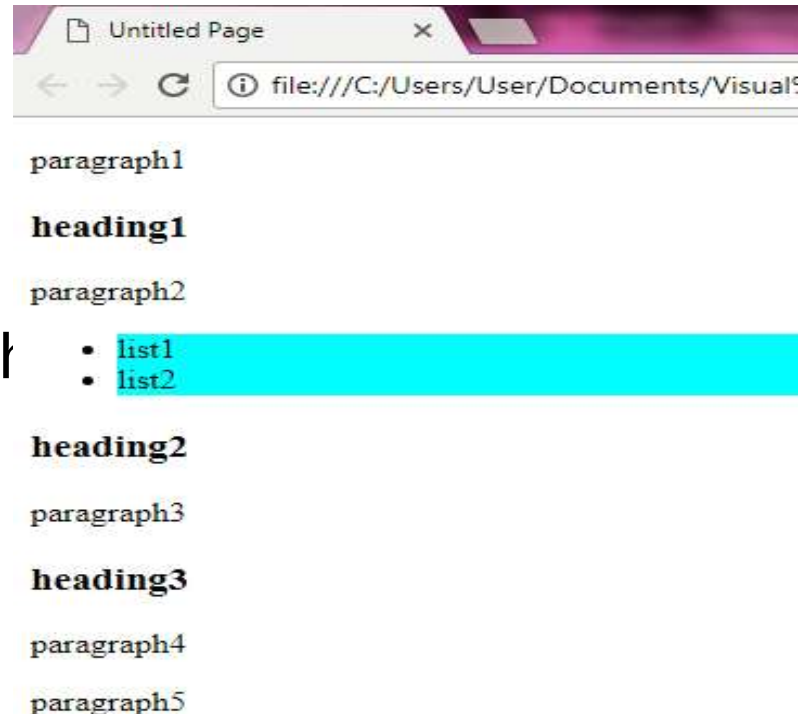


Will select all elements present in a div element

* Selector example

```
<body>
<div>
<p>paragraph1</p>
<h3>heading1</h3>
<p>paragraph2</p>
<ul>
<li>list1</li>
<li>list2</li>
</ul>
</div>
<h3>heading2</h3>
<p>paragraph3 </p><h3>heading3</h3>
<p>paragraph4</p>
<div><p>paragraph5</p></div>
</body>
```

```
div * li
{
    background-color:Aqua;
    color:Black;
}
```



assignment - applying styles to html elements using css

Student registration.html

The form is titled 'Student registration.html' and is set against a light orange background. It contains five input fields and one submit button. The fields are labeled 'Sid:', 'Fname:', 'Lname:', 'States:', and 'Bg:'. The 'States' field is a dropdown menu with a blue triangle icon. The 'Submit' button is a blue rectangle with the word 'Submit' in black text. Green arrows point from the text labels to the corresponding input fields: from 'Sid:' to the first field, from 'Fname:' to the second, from 'Lname:' to the third, from 'States:' to the fourth, and from 'Bg:' to the fifth. A green arrow also points from the 'Submit' button to the text 'Apply style for submit Using Element with attribute selector (bgcolor=aqua,color=red)'.

Apply same style for sid,
Lname using id selector
(bgcolor=green,color=red)

Apply same style for Fname,bg
Using class selector
(bgcolor=blue,color=white)

Apply style for states Using
Element selector
(bgcolor=orange,color=#45ff12)

Apply style for submit Using
Element with attribute selector
(bgcolor=aqua,color=red)

student registration.html

```
myappstyles.css welcome.html* X
Client Objects & Events
<html>
<head>
<title>student registration</title>
</head>
<body>
Sid:<input type="text" id="tbSid"/><br />
Fn:<input type="text" id="tbFn"/><br/>
Ln:<input type="text" id="tbLn"/><br/>

STATES:<select id="ddlstates">
    <option>--select--</option>
    <option>-karnataka</option>
    <option>-kerala</option>
    <option>andrapradesh</option>
</select><br />
bg:<input type="text" id="Tbbg"/><br/>
<input type="submit" id="btnsubmit" value="submit"/>
</body>
</html>
```

adding styles in .css file

```
body
{
    background-color:Yellow;
}
#tbSid,#tbLn
{
    background-color:Green;color:Red
}
.myclass
{
    background-color:Blue;
    color:White;
}
select
{
    background-color:Orange;
    color:#45ff12
}
input[type=submit]
{
    background-color:Aqua;
    color:Red
}
```

Id selector ←

class selector ←

Element selector ←

Element and Attribute selector ←

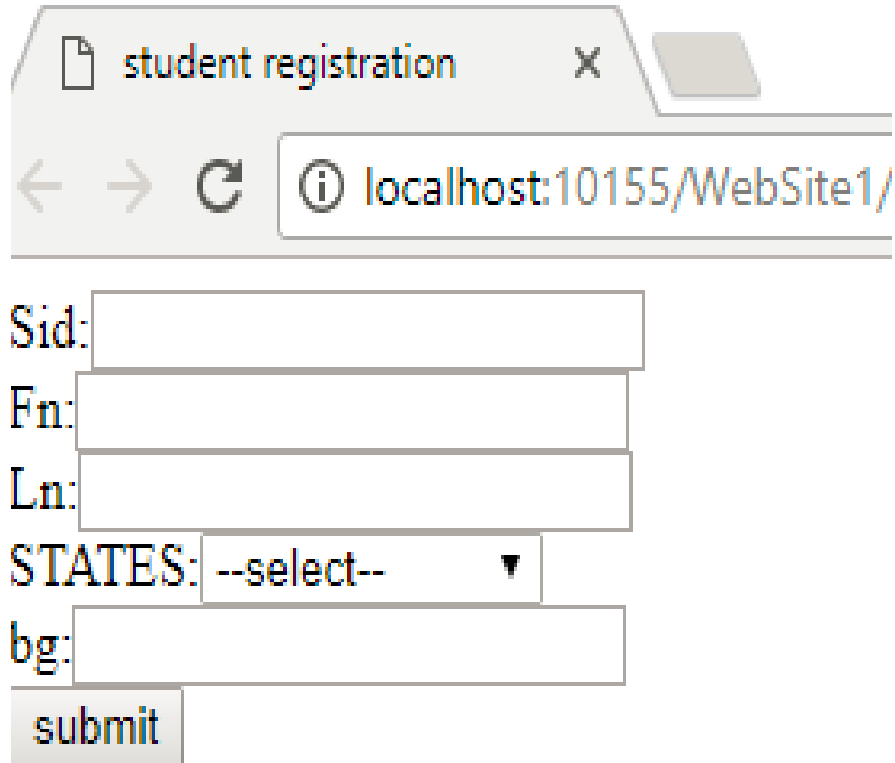
linking html & css file

```
myappstyles.css welcome.html X
Client Objects & Events
<html>
<head>
<title>student registration</title>
<link rel="Stylesheet" href="myappstyles.css" />
</head>
<body>
Sid:<input type="text" id="tbSid"/><br />
Fn:<input type="text" id="tbFn" class="myclass"/><br/>
Ln:<input type="text" id="tbLn"/><br/>

STATES:<select id="ddlstates">
    <option>--select--</option>
    <option>-karnataka</option>
    <option>-kerala</option>
    <option>andrapradesh</option>
</select><br />
bg:<input type="text" id="Tbbg" class="myclass"/><br/>
<input type="submit" id="btnsubmit" value="submit"/>
</body>
</html>
```

Final Output

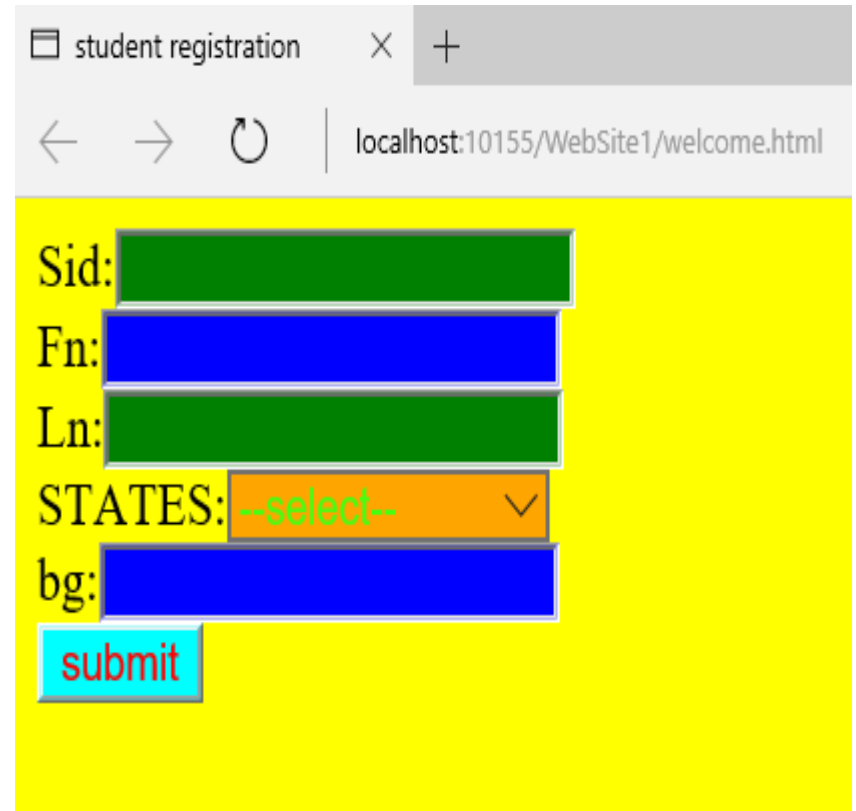
Before linking .css file to html



A screenshot of a web browser window with the title 'student registration'. The address bar shows 'localhost:10155/WebSite1/'. The form contains the following elements:

- Sid:
- Fn:
- Ln:
- STATES:
- bg:
- submit

After linking .css file to html



A screenshot of a web browser window with the title 'student registration'. The address bar shows 'localhost:10155/WebSite1/welcome.html'. The form is styled with a yellow background and has the following elements:

- Sid:
- Fn:
- Ln:
- STATES:
- bg:
- submit

HTML-CSS Assignment

Login.html

EMAIL:

PASSWORD:

login

[New Registration?](#)



Registration.html

First Name:

Last Name:

Age:

Email:

Gender:

☐ Male ☐ Female ☐ Other

state:

Bg:

Qualification: ☐ SSLC ☐ Degree ☐ PG

SUBMIT

Requirements:

First Name:- Class selector(background color-green)

Last Name: Class selector(background color-yellow)

Age,Email: Id selector(background color-aqua)

State: Element selector(background color-blue,color-yellow)

Bg: Id selector (color-Maroon)

Submit button: Element with attribute selector



Execute all code and do
all Exercises at Home

End of
DAY 2